MessFinder: A Web-Based Student Accommodation (Mess) Finding System

Capstone Project Proposal — Part I

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Department of [Your Department] University Name¿University Name

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Abstract

Students relocating to a new university town often struggle to find suitable, affordable, and verified shared accommodation (locally known as a *mess*). Existing solutions rely on word-of-mouth, scattered social-media posts, or unreliable local agents, leading to wasted time, miscommunication, and even fraud. **MessFinder** is a web-based platform that connects *Students* searching for accommodation with *Mess Owners* who list available seats, while *University Proctors* and *Administrators* provide oversight, student-welfare support, and content moderation. This document presents the initial project proposal: problem statement, objectives, scope, development methodology, schedule with Gantt chart, UML/structural diagrams, and an initial working prototype implementing the core domain classes.

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1 Introduction

Finding safe, affordable housing near campus is one of the most common difficulties faced by new and transferring university students. At many universities, including [University Name], students currently search for a mess through informal Facebook groups, senior students' recommendations, or by physically visiting multiple localities — a process that is time-consuming, unstructured, and lacks transparency on pricing, facilities, or seat availability. **MessFinder** aims to digitize and streamline this process through a structured platform with search/filter tools, a recommendation ("Best Match") engine, side-by-side comparison, verified owner listings, reviews, and a student-welfare reporting channel supervised by university proctors.

2 Problem Statement

- Students lack a centralized, searchable directory of available mess seats near campus.
- There is no standardized way to compare rent, facilities, and seat availability across messes.
- Mess owners have no simple digital channel to advertise vacancies and manage booking requests.
- There is no institutional mechanism to track student accommodation welfare issues (e.g., unsafe conditions, disputes) and route them to the appropriate university authority.
- Trust and verification of listings and owners is currently absent.

3 Objectives

1. Design and implement a platform where students can search, filter, and compare mess listings.
2. Provide a recommendation feature (*Best Match*) that ranks listings against a student's budget and preferences.
3. Allow verified mess owners to list, update, and manage seat availability and booking requests.
4. Enable students to submit reviews/ratings and file welfare reports.
5. Provide university proctors a dashboard to review and resolve welfare reports.
6. Provide administrators tools to moderate listings, manage users, and view system analytics.

4 Scope of the Project

4.1 In Scope

- Student, Mess Owner, University Proctor, and Administrator roles with authentication.
- Mess listing CRUD operations, search/filter, best-match ranking, and comparison.
- Booking request workflow (request → approve/reject → seat update).
- Review and rating system tied to each mess listing.
- Welfare report submission and proctor case-handling workflow.
- Administrative moderation and basic analytics.

4.2 Out of Scope (for this phase)

- Online rent payment / payment gateway integration.
- Native mobile applications (web-responsive only in this phase).
- Real-time chat between students and owners.

5 Proposed Solution / System Overview

MessFinder will be delivered as a responsive web application. Students register and search the mess directory using filters such as budget, location, and seat availability; the system also computes a ranked *Best Match* list. Mess owners register and list their properties, and respond to booking requests submitted by students. University proctors monitor and resolve student welfare reports, and administrators oversee the platform's integrity through user management, listing moderation, and analytics dashboards.

6 Tools and Technologies

- **Frontend:** HTML5, CSS3, JavaScript (React/Next.js or equivalent)
- **Backend:** Python / Node.js (REST API)
- **Database:** PostgreSQL / MySQL
- **Version Control:** Git & GitHub
- **Diagrams:** UML (Class, Use Case, Activity, Object), DFD, ER Diagram — produced with Graphviz
- **Documentation:** LaTeX
- **Prototype Language:** Python (console-based OOP prototype, Section 10)

7 Software Process Model

This project follows the **Incremental Development Model**. Core domain functionality (user roles, mess listing, booking workflow) is built first as a working increment (this proposal's prototype), after which each subsequent increment adds features such as recommendation logic, reviews, welfare reporting, and administrative dashboards. This model is chosen because requirements are reasonably well understood at a high level, but individual modules (e.g., the best-match ranking algorithm) are expected to be refined iteratively based on feedback, which an incremental — rather than a strict waterfall — approach accommodates well.

8 Project Schedule, Milestones & Deliverables

8.1 Phases and Milestones

#	Phase	Milestone	Duration
1	Requirement Analysis	Finalized requirements & scope document	7 days
2	Design	Completed UML, DFD, ER diagrams; DB schema	11 days
3	Prototype / Implementation	Working prototype with ≥5 core classes	14 days
4	Testing	Unit & integration test reports	6 days
5	Deployment & Documentation	Final report, GitHub submission	4 days

8.2 Deliverables

- D1: Project Proposal document (this document, PDF via LaTeX)
- D2: UML Diagrams (Class, Use Case, Activity, Object), DFD, ER Diagram
- D3: Gantt chart and schedule
- D4: Working prototype source code (≥5 classes implemented)
- D5: Test report
- D6: Final GitHub repository with README

8.3 Gantt Chart

9 System Diagrams

9.1 Data Flow Diagram (Level 1)

9.2 Use Case Diagram

9.3 Activity Diagram — Booking Workflow

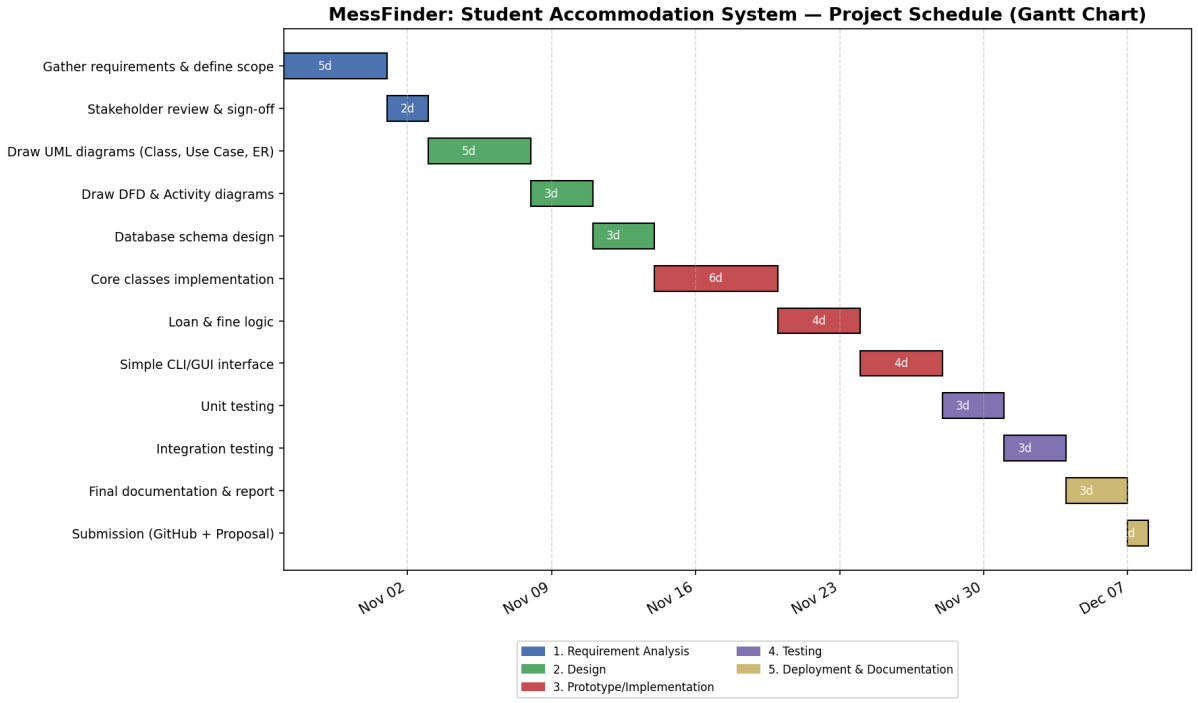


Figure 1: Project schedule Gantt chart

9.4 Entity Relationship Diagram

9.5 Class Diagram

9.6 Object Diagram

10 Initial Prototype

An initial console-based prototype has been implemented in Python, exercising seven classes directly derived from the Class Diagram in Section 7.5: `User` (abstract base), `Room`, `Mess`, `BookingRequest`, `Review`, `Student`, and `MessOwner` (with `Proctor`, `Administrator`, and `WelfareReport` included for completeness). The `Mess`–`Room` aggregation, and the attribute set on `Mess` (area, university, distance, gender, facilities, verification status), were modelled to mirror the data structure already used in the author's own front-end implementation of this system (React/Vite, https://github.com/rashsed60cyber/mess-finding-system), so that this prototype's OOP design can act as the backend data layer for that application. The prototype demonstrates the full workflow: registration, mess search and best-match ranking, comparison, booking request and owner approval, review submission, welfare report filing, and proctor/administrator actions. Source code: `prototype/models.py` and `prototype/main.py` in the accompanying GitHub repository.

11 Expected Outcome

By the end of this capstone project, MessFinder is expected to be a functional web application that meaningfully reduces the time and uncertainty students face when searching for accommodation, while giving mess owners a simple digital management tool and giving the university a structured channel to monitor student welfare concerns related to

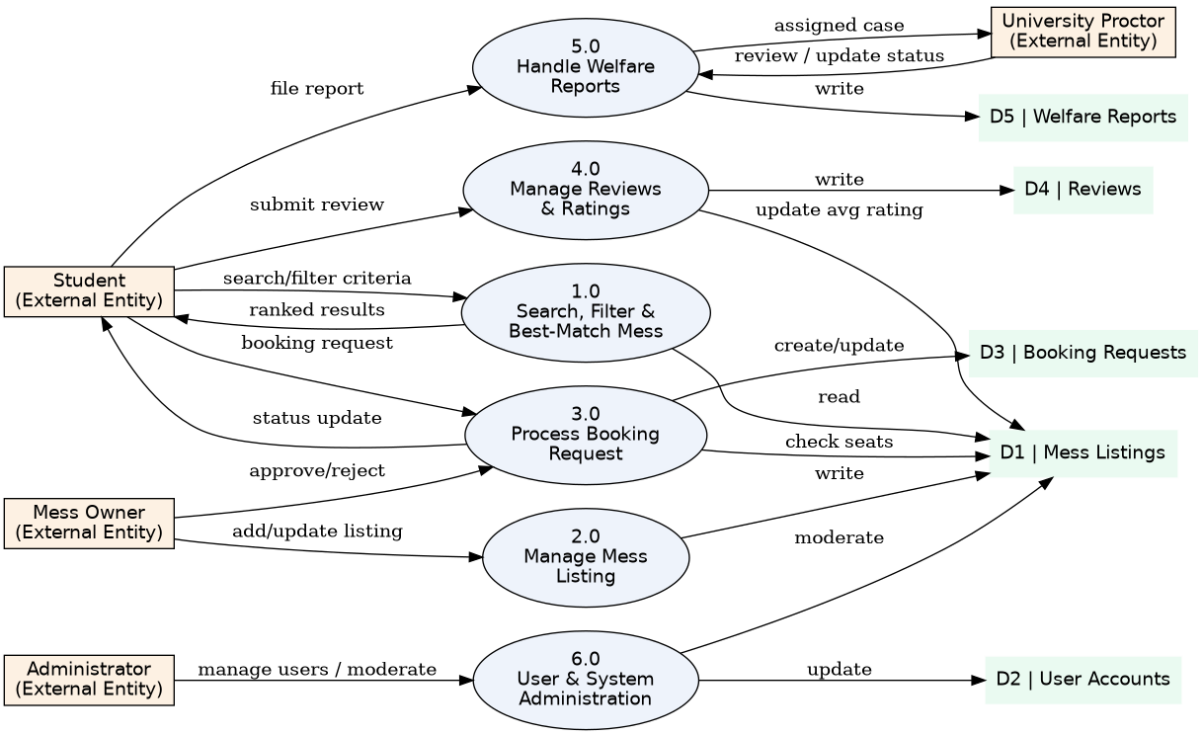


Figure 2: Data Flow Diagram of MessFinder

housing.

12 Repository

Proposal, diagrams, and OOP prototype: https://github.com/<your-username>/<your-repo-name>
Live front-end implementation (React/Vite): https://github.com/rashsed60cyber/mess-finding-system
Deployed demo: https://rashsed60cyber.github.io/mess-finding-system/

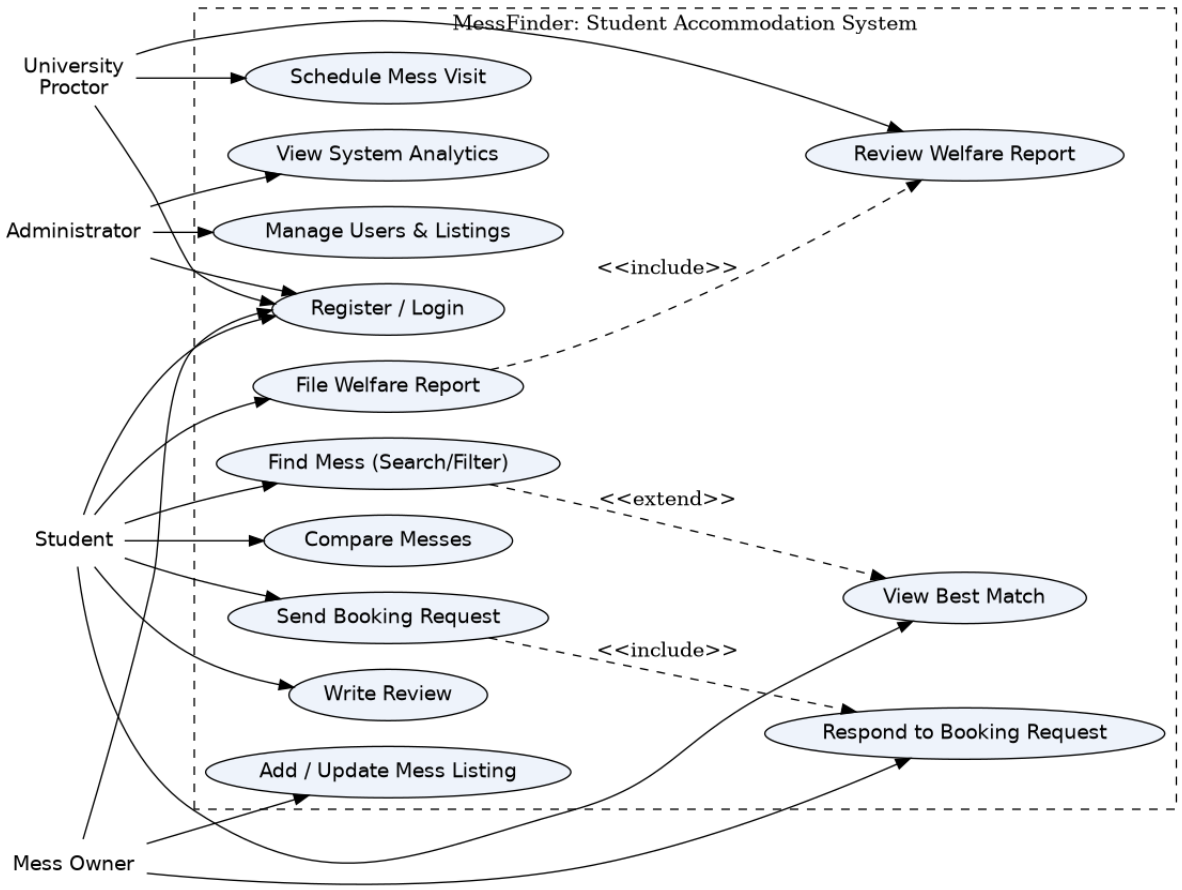


Figure 3: Use Case Diagram — Student, Mess Owner, Proctor, Administrator

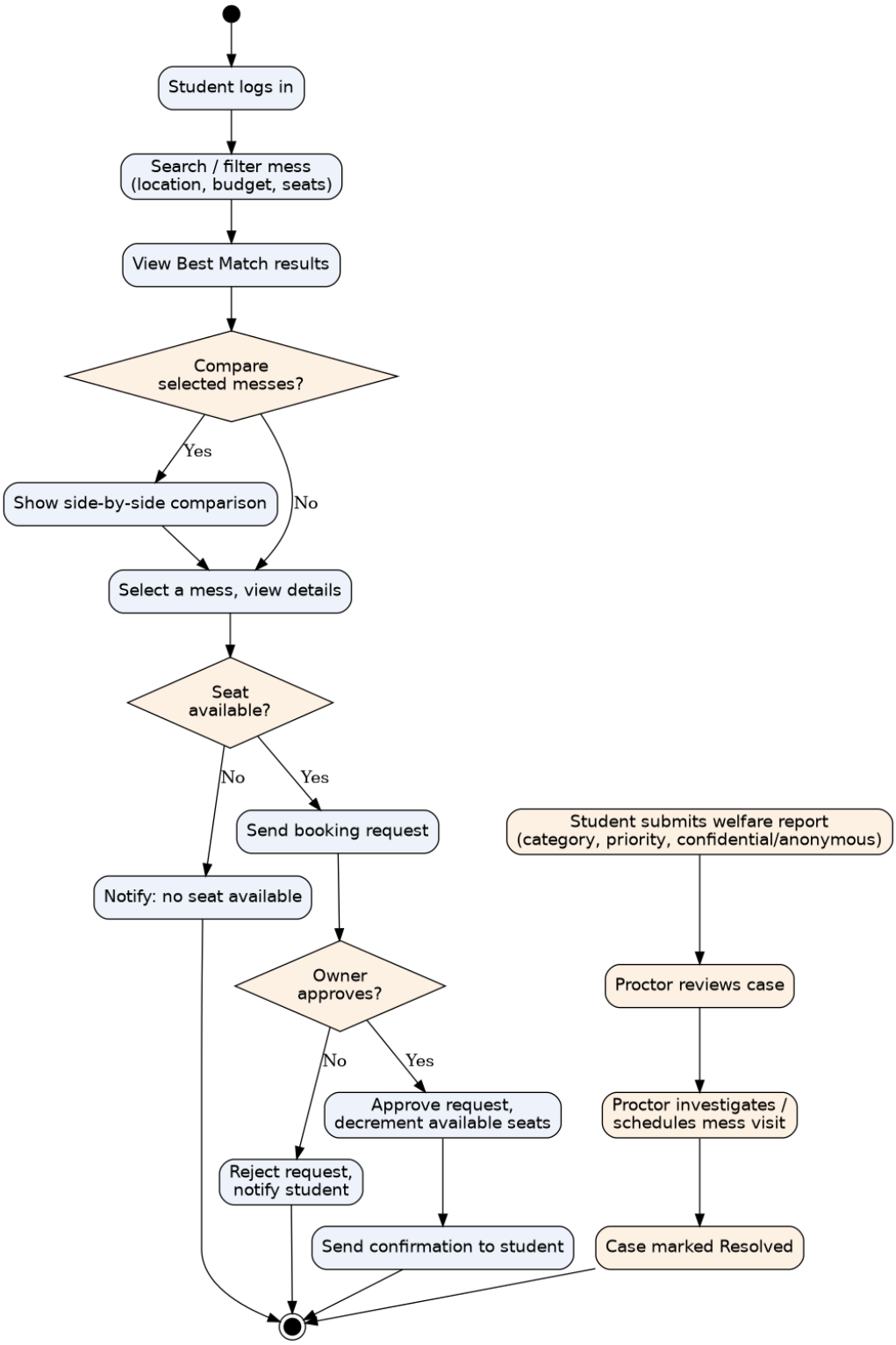


Figure 4: Activity Diagram for the mess search-to-booking workflow

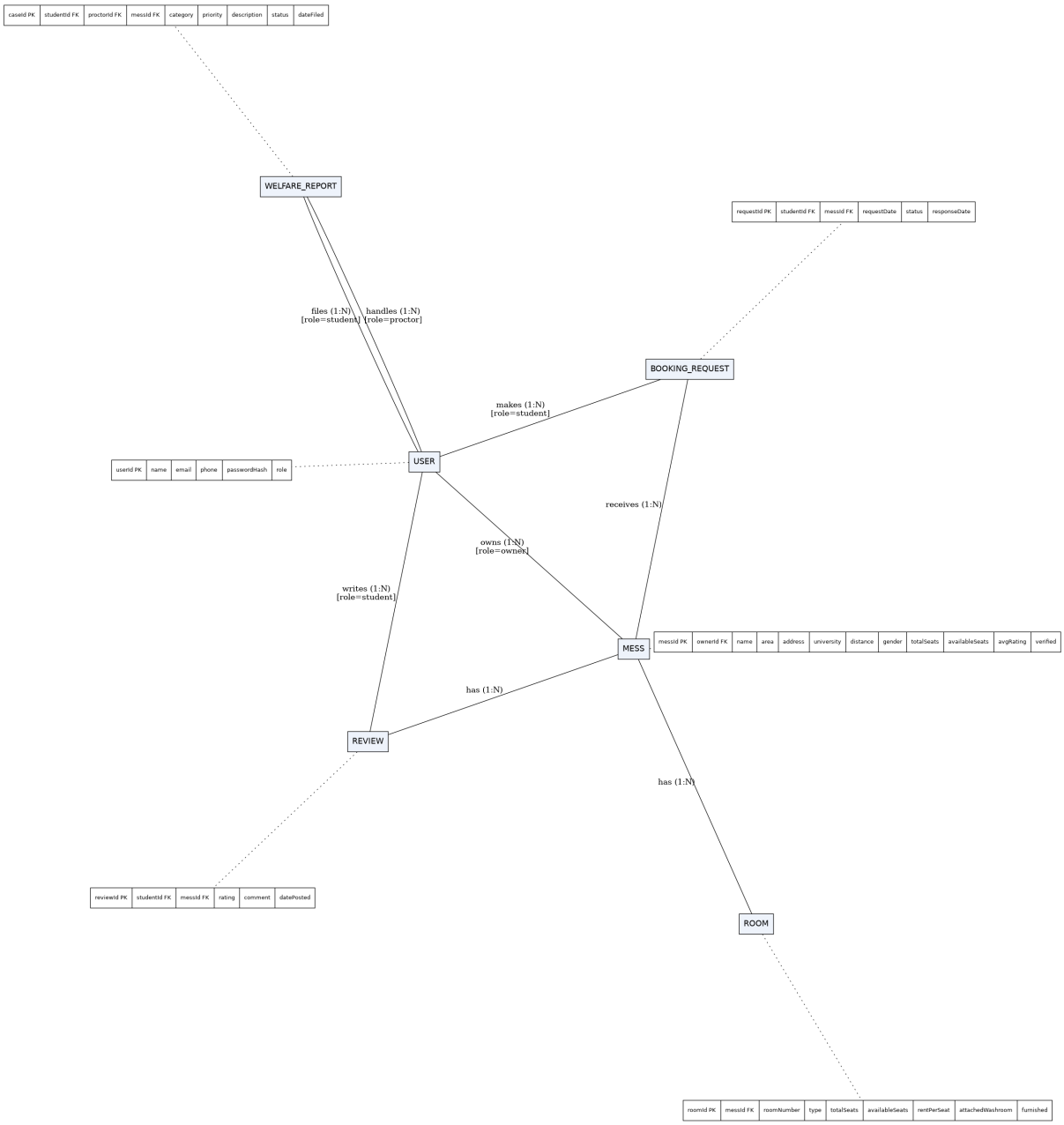


Figure 5: Entity Relationship Diagram

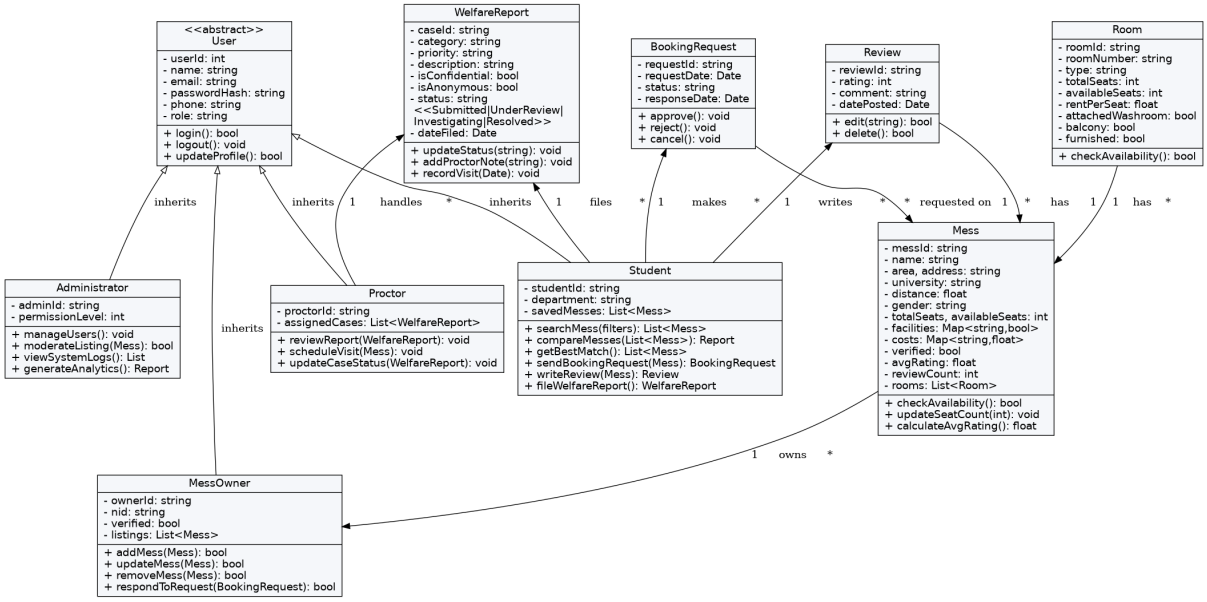


Figure 6: Class Diagram

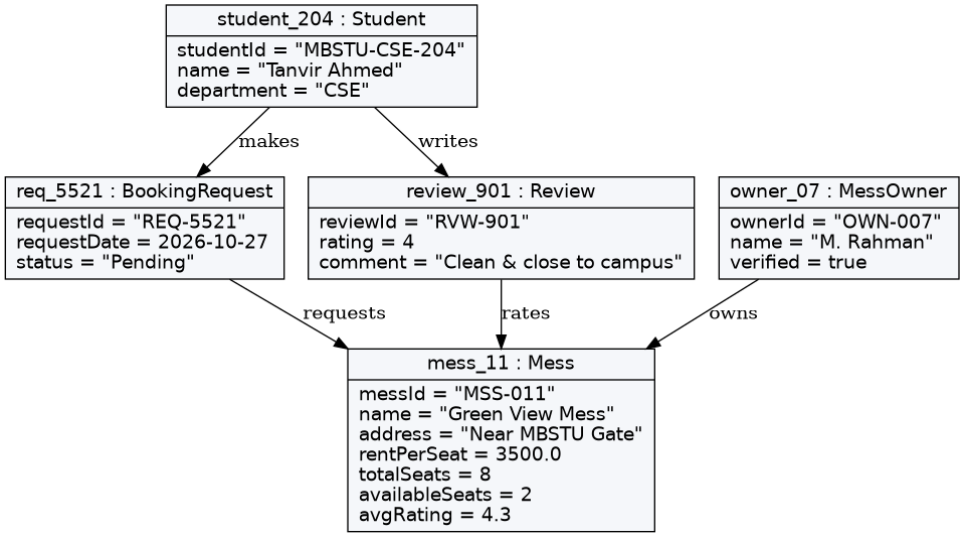


Figure 7: Object Diagram — a sample runtime snapshot